



Day 5: Visualizations & Report Building

Today we transform business data into stories. This is the day where dashboards finally come alive.

The Amazon Data Analyst Mission

Amazon E-Commerce Case

Imagine you are working as a Data Analyst at Amazon. Management presents you with 10,000 transaction rows and asks:

-  Which geographic region has the highest overall sales?
-  Which product category yields the highest profit margins?
-  Which month shows the best performance trend?
-  Which payment options are trending?

Dashboard Purpose

Can executives answer these questions quickly by looking at a flat spreadsheet table of 10,000 rows?

No. Reading row-by-row is impossible at scale.

"Today we learn how dashboards answer business questions visually."

What is Data Visualization?

🎯 Visual Mapping

Data Visualization means representing complex numeric datasets using clear graphical charts and indicators for rapid comprehension and decision-making.

A dashboard provides an consolidated view of operational health at a single glance.

🚗 The Dashboard Analogy

"A business dashboard is like a car speedometer. You don't read raw engine code while driving. You simply glance at the dial."



120 KM/H

SPEED INDICATOR

Bar Chart: Category Comparison

Best For: Category Comparison

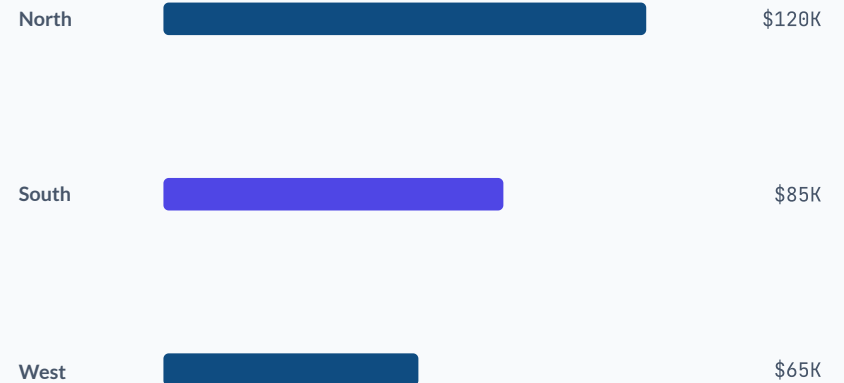
Bar charts are highly effective for comparing metric sizes across distinct, discrete categories.

Common Use Cases:

- 📍 Sales Revenue by Region
- 💰 Profit margins by Category
- 📅 Order counts by State

"Clustered bar charts are the most commonly used, easy-to-read business visuals."

📊 Sales Revenue by Region



Line Chart: Time Trends

Best For: Time Series Trends

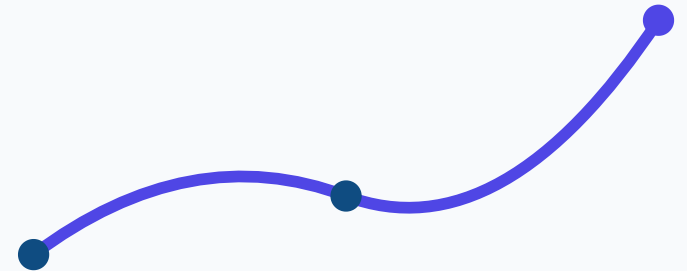
Line charts connect sequential data points with straight lines, making them the industry standard for showing changes over time.

Common Use Cases:

- 📅 Monthly sales trends
- 🕒 Daily order spikes
- 📈 Annual profit margins

"Line charts are the best way to visualize growth patterns and seasonal declines."

📈 Monthly Sales Trend (Q1 - Q4)



Pie Chart: Percentage Share

Best For: Part-to-Whole Share

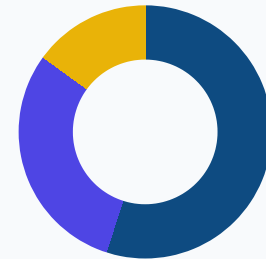
Pie charts show how much individual categories contribute to a single total metric, represented as percentage wedges.

Common Use Cases:

- Category market shares
- Regional revenue contribution
- Customer segment analysis

"⚠ Warning: Avoid using pie charts for columns containing more than 3 or 4 categories."

Category Contribution Share



■ Technology (55%) ■ Office (30%) ■ Furniture (15%)

KPI Cards & Interactive Slicers

KPI Card Visuals

Cards display a single, high-priority metric in a large, easy-to-read format. They are the first element executive users look for on a report.

Primary KPIs:

-  **Total Sales:** Overall revenue.
-  **Total Profit:** Net profit.
-  **Total Orders:** Transaction count.

Interactive Slicers

Slicers are on-screen filter controls that let users dynamically segment the data shown across all dashboard visuals.

Example: An Amazon regional manager selecting ****South Region**** to filter and focus solely on Southern sales performance.

Visual Formatting & Layout Design

✓ Layout Best Practices

- ✓ **Executive KPI Cards:** Place at the top of the page.
- ✓ **Trends & Categories:** Place in the center body.
- ✓ **Interactive Slicers:** Align along the left or right margins.
- ✓ **Design Principles:** Use clean, minimalist, high-contrast themes.

⚠ Common Mistakes

- ✗ **Color Overuse:** Using too many vibrant colors.
- ✗ **Tiny Fonts:** Hard-to-read text values.
- ✗ **Visual Overload:** Crowding the canvas with too many charts.
- ✗ **No Structure:** Messy, unaligned layouts.

“A good dashboard explains business trends quickly—it should never confuse the user.”

How to Build Your Dashboard



Steps 1 to 4: Set Up

Step 1: Import Data: Go to Home → Get Data → Text/CSV and load `Sample - Superstore.csv`.

Step 2: Build KPI Cards: Create KPI cards at the top for Total Sales, Total Profit, and Total Orders.

Step 3: Compare Regions: Add a Bar Chart to compare overall Sales by Region.

Step 4: Visualize Trends: Add a Line Chart showing monthly sales trends over time.



Steps 5 to 7: Polish

Step 5: Show Contributions: Add a Pie Chart showing Sales by Category contribution.

Step 6: Add Slicer: Place a Region Slicer on the left sidebar to enable interactive filtering.

Step 7: Format & Clean: Standardize titles, enable data labels, and verify layout alignment.

Interactive Dashboard Workspace

Interactive Slicer Filter

ALL REGIONS

Select: All Regions

Select: North Region Only

Select: South Region Only

TOTAL SALES KPI

\$270K

TOTAL PROFIT KPI

\$45K

REGIONAL BAR CHART (SALES)

North

South

West

Exam Guidelines & Sector Use Cases

Exam Study Guide

-  **Bar Chart:** Best visual choice for categorical comparisons.
-  **Line Chart:** Best visual choice for chronological trend analysis.
-  **Pie Chart:** Best for percentage share contribution of 3 categories or fewer.
-  **Card Visual:** Best for highlighting high-level KPI values.
-  **Slicer:** Used for on-screen interactive page filtering.

Sector Implementations

-  **Retail Analytics:** Sales margins, regional volume, and category returns.
-  **Banking:** Loan allocations, customer growth, and monthly revenue.
-  **Healthcare:** Patient admission counts, hospital occupancy, and clinical trend lines.
-  **HR:** Employee turnover, headcount by department, and training completion rates.

Dashboard Completion

“ Trainer Lines

“Today, you transformed flat rows of data into active business insights. This is where analytics becomes storytelling.”

“Data alone is just numbers... Visualization turns numbers into decisions.”

★ Interactive Engagement Check

Let's check our visual choice decisions before wrapping up:

- ❓ "If you were the company CEO, which KPI card would you check first?"
- ❓ "Which visual chart type is best suited for showing monthly trends?"
- ❓ "Who can build the cleanest, most minimalist dashboard layout?"